

Summary

Nowadays, natural parks serve for both conservation of the environment and human entertainment. However, human interaction with the environment must be restricted not to lead to unregulated illegal animal hunting, which can be detrimental to the park's ecosystem and world's animal diversity.

Our goal was to create an objective, measurable definition of a national park protection level and create a model for human and technological resources allocation.

First we divided the natural park into 15 zones, based mainly on rare black rhinos allocation, but also considering landscape type, separating salt pans to separate zones.

For the purpose of model creation we considered – **environmental factors**:

- rain level, which reducing wildfire probability
- rhino population level in each zone, increasing poach probability
- animal population level in each zone, increasing poach probability
- number of waterholes, increasing poach probability;

human infrastructure factors:

- road length, with longer roads leading to increased poach probability
- main camp presence, decreasing poach probability;

and **previous events**, such as fires in the last 25 years.

Our model accounted for these factors to minimize unneutralized poach accidents and wildfire events by allocation of patrols and drones in the most effective way.

We also explored the change in our modelling for changes in financial restrictions and personnel limitations, and it shows that our model's resources allocation can depend on these characteristics changing.

Our model was then generalized to Yellowstone national park in USA, and Komodo national park in Indonesia, concluding that adjustment of numerical variable scores will allow us to measure protection levels in other national parks, however it will work the best for parks with main protection problems of poaching and wildfire.

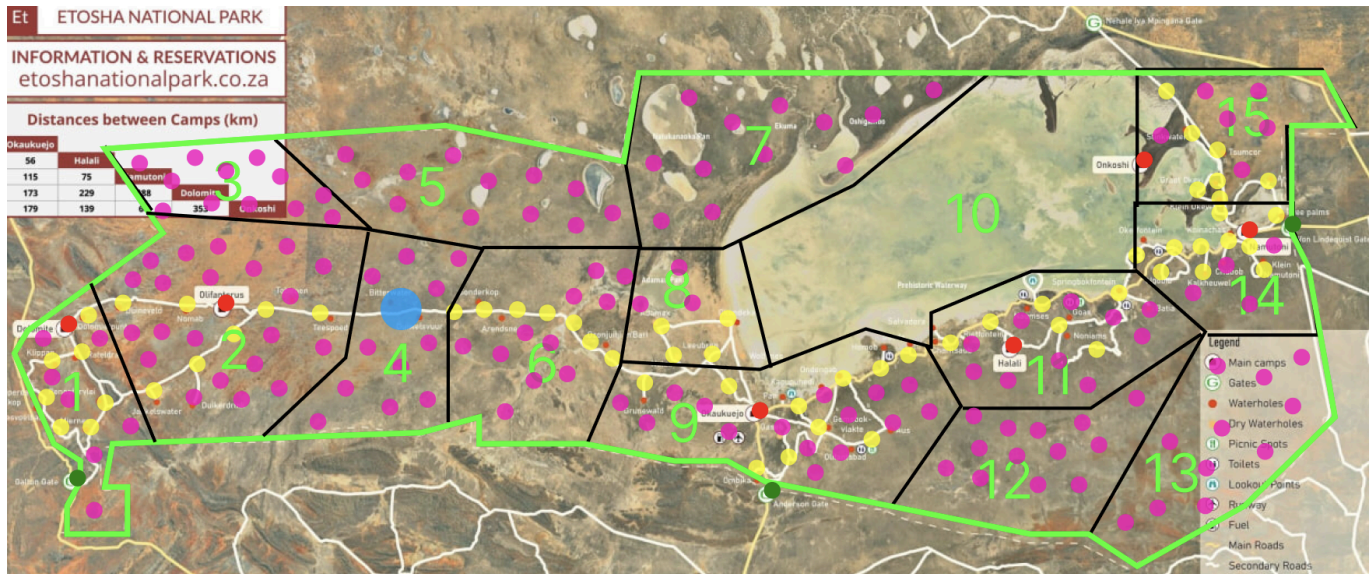
Finally, we analysed our model in terms of generalizability and effectiveness, and found it to be sufficiently good for Etosha park protection modelling.

Letter to the IMMC

Dear Directors of IMMC,

Our team developed a proposal for resource allocation in Etosha National Park, that we are very excited to share with you!

From our summary sheet, we hope that you are already familiar with our modeling method and criteria that we accounted for. We identified where patrols, drones and fire units should be placed on the map below (drones are pink, patrols are yellow, fire unit is blue):



The suggested solution allows the park to effectively fight the problems of poaching and wildfire damaging the park flora and fauna by minimizing the probability of uninterrupted poaching or of a fire that was not extinguished in time.

We believe that the implementation of such a protection system will help to establish the welfare of the park, its visitors and inhabitants.

Kindest regards,
 Team #2026009

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1. Introduction

1.1. Context

Historically, national parks were designed as a place for public hunting. Now, the concept of natural parks significantly changed, as those aim for conservation and public enjoyment. (Sara Litke, 1998)

Etosha National Park, located in Namibia, is a unique area, as multiple landscapes are present there, including a large salt pan, that can be seen from space; the abundance of various animal species including both common (such as elephants) and rare (like endangered black rhinos) ones. However, like any natural environment it faces a variety of challenges. We identified black rhino (and other animals) poaching and wildfire protection to be the most important for the natural park sustainability.

1.2. Modelling objectives

Our goal was to create a model for protection planning for Etosha National Park. This model must rely on numerical data and the definition of protection in a quantitative, objective way. Our plan for model development, including relevant criteria is presented in Figure 1.

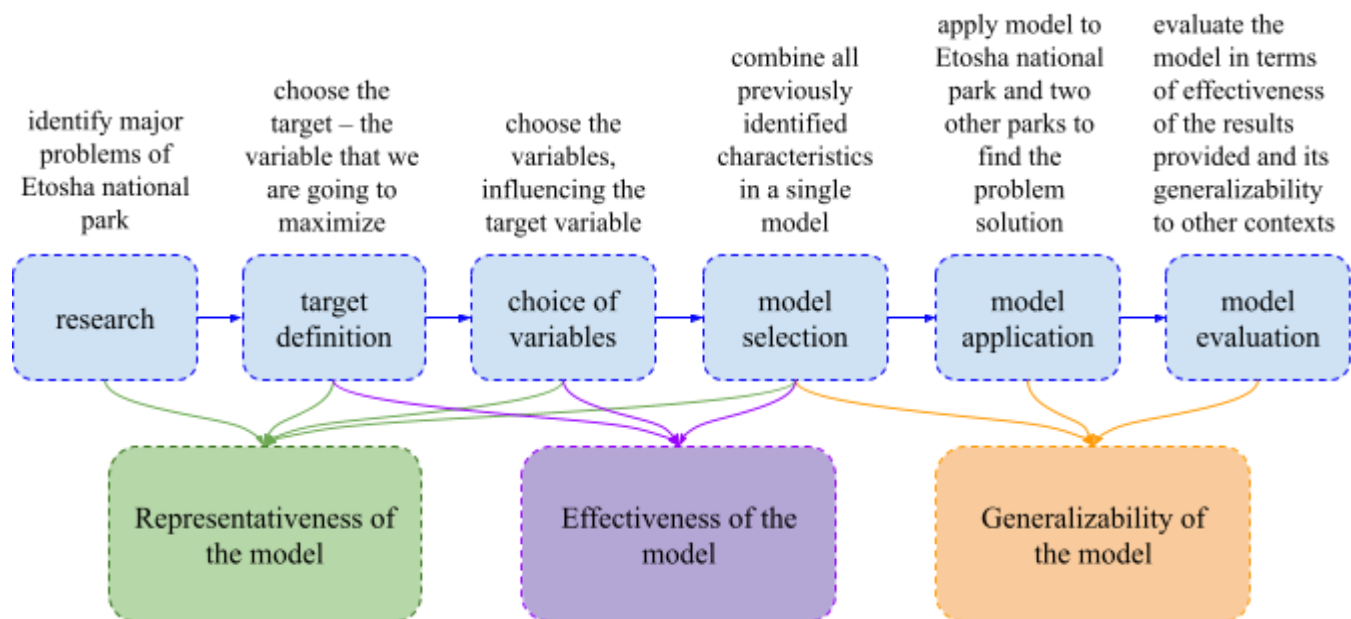


Figure 1. Modelling plan and criteria of success.

1. Assumptions and variables

2.1. Key definitions

When constructing our model it was important to have a clear definition of Protection level, based on the following considerations:

- Protection level must be represented as a variable, which can be calculated as a number in a given condition of the park
- It should account for a variety of factors, including geographical features, characteristics of the landscape and last year's events.

Finally, we decided to calculate the protection level as a probability of a harmful event not happening, or happening and being neutralized. Or, alternatively, $1 - \text{probability of a harmful event happening}$.

2.2. Variable definition

Zones general features		
area[i]	$\in \mathbb{N}$	Area of the zone i
gate[i]	$\in \{0; 1\}$	Gate presence/absense in the zone i
fence [i]	$\in [0; 1]$	$\frac{\text{fence perimeter in zone } i}{\text{zone perimeter in zone } i}$
water[i]	$\in \mathbb{N}$	The number of waterholes in a zone
rhino[i]	$\in \{0; 1; 2\}$	Rhino population level in zone i: 2 – if seen, 1 – seen nearby, 0 – minimal probability
animal[i]	$\in \{0; 1; 2\}$	Animal population level in zone i: 2 – if frequently seen, 1 – seen not frequently/seen nearby, 0 – minimal probability
rain[i]	$\in [0; 1]$	Rain level in the zone
fire[i]	$\in \mathbb{R}, \geq 0$	Average number of fire accidents in past 25 years
road[i]	$\in [0; 1]$	Road length/area - [0; 1]
maincamp[i]	$\in \{0; 1\}$	Main camp - 0/1
neighbours[i]	$\in \mathbb{N}$	Neighbours (zones)
General force features		
patrol_range	80	km ²
drone_range	128	km ²
patrol_coverage_zone[i]	$\in \{0; 1\}$	Patrol coverage zone / area of the zone
drone_coverage_zone[i]	$\in \{0; 1\}$	Drone coverage zone / area op the zone
Poach-related variables		
poach_detect [i]	$\in \mathbb{R}, \geq 0$	Zone poach detection
patrol_access [i]	$\in \mathbb{R}, \geq 0$	Patrol access zone
poach_prevent[i]	$\in \mathbb{R}, \geq 0$	Poach prevent zone – the probability of stopping an act of poach in zone i

poach_prob[i]	$\in \{0; 1\}$	Poach probability in zone i
Wildfire-related variables		
		Analogical to poach-related

Table 1. Variable definition

2.3. Assumptions

Assumption #1: Probability of wildfire in salt pans is equal to 0.

Explanation: Salt requires extremely high temperatures to melt and does not react with the air. For this reason salt pans are considered to be non-inflamable.

Assumption #2: If a protection unit is located in the area i, we consider it can identify and neutralize the threat.

Explanation: As it is impossible to account for all the movements of the units within the zone, we assume that if the unit is located wherever in the zone it can cope with the problems in that zone.

2. Mathematical model

The model consists of two sequential stages: poaching prevention and wildfire mitigation. In the first stage we analyze strategically valuable placement for drones and patrols designed to detect and prevent poaching incidents, as this is one of the major problems in Etosha National Park. After those are positioned, we optimize the placement of additional drones and fire units to minimize wildfire risks. The resulting network minimizes the risks of both poaching attacks and wildfire occurrence. The probability of an incident occurring consists of two multiplicative terms: the probability of an incident occurring, and the probability of it not being stopped given that it occurred. In our model, the first term is given by `poach_prob` and `fire_prob`, and the second term by `1 - poach_prevent` and `1 - fire_prevent`, for poaching and wildfire respectively. Our objective is to minimize these products. However, as Etosha National Park is a vast and heterogeneous area, we divide the reserve into 15 zones in order to refine our calculations and better approximate real-world conditions. The zoning is based primarily on rhino density, reflecting the central role of poaching prevention in our model.

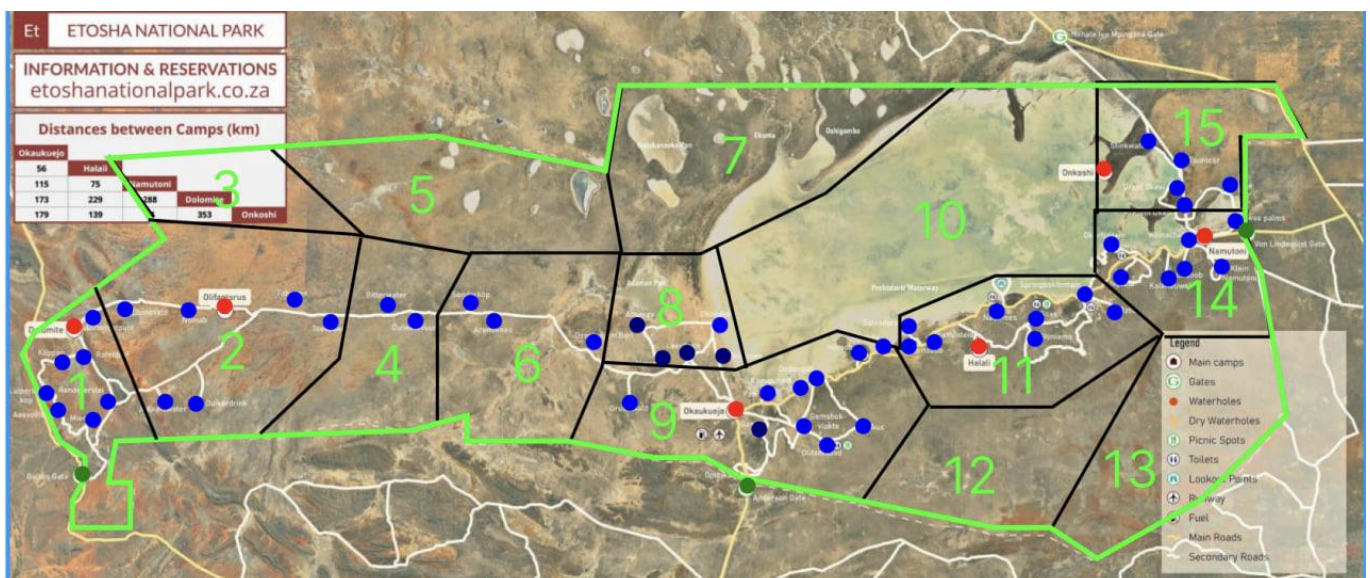


Figure 1: zones division

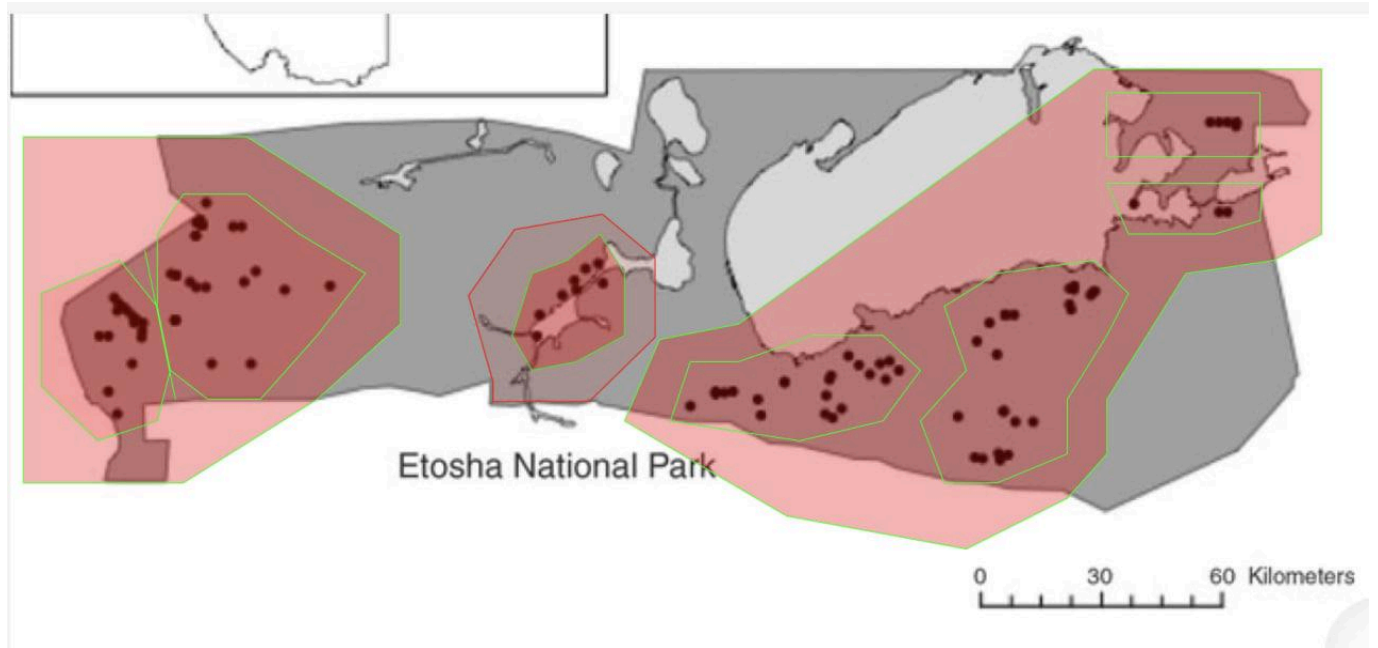


Figure 2: black rhinoceros population

According to data from research articles, we have specified the values for each zone these features:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
area[i]	1408	3219	1207	1811	2213	2011	2625	905	2715	4325	1609	1710	2011	1257	1106
gate[i]	1	0	0	0	0	0	0	0	1	0	0	0	0	1	0
fence[i]	0.7	0.3	0.4	0.25	0.4	0.2	0.45	0	0.4	0.1	0	0.2	0.6	0.25	0.55
water[i]	7	5	0	2	0	3	0	1	9	0	8	0	0	7	5
rhino[i]	2	2	1	0	0	2	0	0	2	0	2	2	0	2	2
animal[i]	2	2	1	1	0	2	0	2	2	0	2	2	0	2	2
rain[i]	0	0.25	0.4	0.4	0.5	0.3	0.4	0.2	0.6	0	0.75	0.82	1	0.85	0.85
fire[i]	1	6	7.5	4	5	3	1.5	1	2.5	0	3	4	4	2	0.5
road[i]	1	0.6	0	0.3	0	0.4	0	0.5	0.8	0	0.75	0	0	0.8	0.6
main camp[i]	1	1	0	0	0	0	0	0	1	0	1	0	0	1	1
neighbours[i]		1; 3; 2 4	2; 5 6	2; 5; 6	3; 4; 6; 7	4; 5; 8; 9	5; 8; 10	6; 7; 9; 10	6; 8; 10; 11;	7; 8; 9; 11; 14; 15	9; 10; 12; 14	9; 11; 12; 13	12; 14	10; 11; 13; 15	10; 14

Table 2. Zone features values.

3.1. Poaching prevention stage

The poaching prevention stage allocates ground patrol units and drone surveillance across the 15 zones to minimize the expected poached area, defined as the zone area weighted by its residual poaching risk after prevention is applied. Formally, the objective of this stage is:

$$\text{minimize sum over } i \text{ of } \text{poachRisk}[i] * (1 - \text{Prevent}[i]) * \text{area}[i]$$

The poaching risk for each zone is computed as a composite score that reflects both the attractiveness of the zone to poachers and their operational ease of access. We model it as:

$$\text{poachRisk}[i] = (W2 * \text{rhino}[i] + W1 * \text{animal}[i]) * (\text{road}[i] + 1) * (\text{water}[i] + 1) * (2 - W3 * \text{maincamp}[i])$$

The first factor looks at the biological value of the zone. It gives more weight to the presence of rhinos ($W2 = 3$) than to the presence of other animals ($W1 = 1$). This is because rhinos are more likely to be poached because their horns are worth a lot of money. The terms for road and water access come in as multiples instead of as sums. This is because a zone that is both road-accessible and water-rich is much easier for poachers to work in than a zone that only has one of these features. By adding 1 to each term, zones with no road or water access still have a baseline risk level instead of being completely zeroed out, which would be an unrealistic assumption. Finally, the presence of a main camp is reducing the risk score, as there is likely to be a great flow of people; the coefficient $W3 = 0.9$ reflects that camps suppress poaching activity but do not eliminate it.

The model allocates 60 ground patrol units and 100 drone units across the 15 zones, with a maximum of 11 patrol units per zone to enforce operational realism and prevent the trivial solution of concentrating all resources in a single zone. The decision variables $P[i]$ and $D[i]$ represent the number of patrol and drone units assigned to zone i , respectively.

The effective coverage of zone i is defined as the fraction of the zone's area that falls within the operational reach of deployed resources:

$$\text{Coverage}[i] = (\text{patrolRange} * \text{road}[i] * P[i] + \text{droneRange} * D[i]) / \text{area}[i]$$

Ground patrols operate with a range of 80 area units per unit, but their contribution is scaled by road accessibility, since patrols are fundamentally constrained in their mobility by the road network. Drones, with a range of 128 area units per unit, are not subject to this constraint and contribute unconditionally. Coverage is bounded between 0 and 1, as it represents a proportion. This formulation ensures that zones with poor road infrastructure are naturally nudged toward drone-heavy solutions, which reflects realistic operational planning.

Raw coverage does not translate linearly into detection probability, as additional resources in an already well-covered zone yield progressively smaller gains. We therefore model detection using a piecewise linear concave function that approximates diminishing returns:

$$\text{Det}[i] \leq \beta_1 * \text{Coverage}[i]; \text{Det}[i] \leq \beta_1 \text{cp} + \beta_2(\text{Coverage}[i] - \text{cp}); \text{Det}[i] \leq 1$$

The first segment, with slope $\beta_1 = 0.6$, applies when coverage is below the breakpoint $\text{cp} = 0.5$, capturing the rapid gains in detection probability when resources are first deployed. Above this threshold, the slope drops to $\beta_2 = 0.3$, reflecting saturation effects as the zone becomes increasingly monitored. Because the objective minimizes residual risk, the optimizer will naturally drive $\text{Det}[i]$ to the maximum value permitted by these constraints, meaning the piecewise linear upper envelope is tight at optimality. This concave structure is both empirically justified and computationally convenient, as it preserves the linearity of the model without requiring binary variables.

Detection alone is insufficient to prevent poaching: rangers must also be physically present to intercept detected poachers. We therefore define a separate Access variable that captures ground patrol presence in zone i , including the contribution of patrols stationed in neighboring zones through a spatial spillover mechanism:

$$Access[i] \leq P[i]/maxPatrolPerZone + \alpha[i] * \text{sum over } j \text{ in neighbors}(i) \text{ of } P[j]/maxPatrolPerZone$$

The rationale for including spillover is that a patrol unit stationed in an adjacent zone is not confined to that zone; it can respond to incidents across the boundary within a realistic response time. The spillover weight $\alpha[i] = 1 / |\text{neighbors}(i)|$ normalizes the contribution so that a zone surrounded by many neighbors does not receive disproportionate benefit from the same number of neighboring patrols compared to a zone with fewer neighbors. Both the own-zone and neighboring contributions are expressed as fractions of the maximum allowable patrol count, ensuring Access remains bounded in $[0, 1]$.

The overall prevention effectiveness in zone i is the joint outcome of detection and physical access, since an incident can only be stopped if it is both detected and responded to. This requires the product of $Det[i]$ and $Access[i]$, which would introduce a nonlinearity. To preserve the MIP-compatibility of the model, we linearize this product using McCormick envelope constraints:

$$Prevent[i] \leq Det[i]; \quad Prevent[i] \leq Access[i]; \quad Prevent[i] \geq Det[i] + Access[i] - 1$$

These three inequalities define the tightest possible linear relaxation of the bilinear product of two variables bounded in $[0, 1]$. The first two constraints ensure that prevention cannot exceed either of its components, while the third enforces that prevention is not underestimated when both components are simultaneously high. At optimality, $Prevent[i]$ converges to the true product $Det[i] * Access[i]$ for any solution in which both variables are driven to their bounds by the objective, which is the case here since minimizing residual risk rewards maximizing prevention.

Using CPLEX optimising this multi-linear model we get the following placement of our forces:

$$P = [7 \ 7 \ 0 \ 0 \ 0 \ 7 \ 1 \ 4 \ 10 \ 0 \ 5 \ 1 \ 0 \ 11 \ 7];$$

$$D = [7 \ 22 \ 6 \ 0 \ 0 \ 13 \ 0 \ 3 \ 16 \ 0 \ 10 \ 13 \ 0 \ 4 \ 6];$$

3.2. Wildfire prevention stage

The wildfire prevention stage optimizes the deployment of additional fire response units and drone surveillance across the 15 zones, taking the patrol and drone allocations from the poaching prevention stage as fixed inputs. The objective of this stage is to minimize the expected area exposed to unmitigated wildfire risk:

$$\text{minimize sum over } i \text{ of } (1 - rain[i]) * fire_accidents[i] * (1 - fire_prevent_zone[i]) * area[i]$$

Each zone's wildfire risk score takes into account both the terrain's structural fire hazard and the frequency of past fire incidents. The term $(1 - rain[i])$ reduces the contribution of zones where rainfall offers some passive protection because areas with higher precipitation are inherently less likely to sustain fires. Rainfall enters the target as a suppressor. In each zone, the $fire_accidents[i]$ parameter records the historical occurrence of fire events and serves as a proxy for the underlying ignition risk caused by factors such as vegetation type, human activity, and lightning frequency. Together, these two terms characterize a zone-level fire hazard that the model seeks to reduce through resource allocation.

Unlike the poaching risk formulation, wildfire risk does not include a spatial accessibility multiplier of the same complexity, since fire ignition is less dependent on deliberate agent behavior and more on environmental conditions. The risk terms therefore enter the objective linearly and multiplicatively,

consistent with the interpretation that the expected damage in zone i is proportional to its hazard level and the area left unprotected.

Two new decision variables are introduced in this stage: $\text{fire_drones}[i]$, which indicates the number of additional drones assigned to zone i for fire detection, and $\text{fire_unit}[i]$, which is a binary variable that indicates whether a dedicated fire response unit is stationed in zone i . Due to the limited number of specialized wildfire intervention teams in the park, there can only be one fire response unit overall. Additionally, the total budget for drones at this stage is 60 units, with a maximum of 12 drones per zone. This value was investigated empirically: we have tried all possible maximums and the weighted minimum was at the point of 12 pieces per zone. This happens because of local extremums of our function, when the model stop. Importantly, $\text{const_patrols}[i]$ and $\text{const_drones}[i]$, the patrol and drone allocations decided upon in the poaching prevention stage, enter this stage as fixed parameters, signifying the pre-existing surveillance capacity that also serves as a secondary function in fire detection.

Fire detection coverage in each zone aggregates contributions from three sources: the pre-existing drone fleet from the poaching stage, the pre-existing ground patrols from the poaching stage, and the newly allocated fire drones. These are combined as raw coverage quantities rather than proportions of zone area:

$$\text{zone_fire_detection}[i] \leq \text{adj_drone_coverage}[i] + \text{const_drone_coverage}[i] + \text{const_patrol_coverage}[i]; \text{zone_fire_detection}[i] \leq 1$$

where $\text{const_drone_coverage}[i] = \text{drone_range} * \text{const_drones}[i]$, $\text{const_patrol_coverage}[i] = \text{road}[i] * \text{patrol_range} * \text{const_patrols}[i]$, and $\text{adj_drone_coverage}[i] = \text{drone_range} * \text{fire_drones}[i]$. Ground patrol coverage is again scaled by road accessibility for the same reasons discussed in the poaching stage. The detection variable is bounded above by 1, and since the objective rewards maximizing $\text{fire_prevent_zone}[i]$, the optimizer will drive $\text{zone_fire_detection}[i]$ to the minimum of its upper bounds, effectively implementing a capped linear detection function without requiring explicit nonlinear operators. This approach avoids the use of a $\min()$ function, which would break the linearity of the model.

The physical capacity to respond to a detected fire depends on whether a fire response unit is present in the zone or in an immediately neighboring zone. We define $\text{fire_unit_access}[i]$ to capture this, incorporating the same normalized spatial spillover mechanism used in the poaching stage:

$$\begin{aligned} \text{fire_unit_access}[i] &\leq \text{fire_unit}[i] + \alpha[i] * \text{sum over } j \text{ in neighbors}(i) \text{ of } \text{fire_unit}[j]; \\ \text{fire_unit_access}[i] &\geq \text{fire_unit}[i] \end{aligned}$$

The lower bound constraint ensures that if a fire unit is stationed directly in zone i , the access variable reflects this regardless of the spillover term, preventing the optimizer from undervaluing direct deployment. Since $\text{fire_unit}[i]$ is binary, $\text{fire_unit_access}[i]$ takes values in $\{0, 1\}$ at optimality, with intermediate values possible only when neighboring units contribute partial spillover.

Because a fire response unit provides suppression capacity even in zones without a directly assigned unit, we do not wish to model access as a hard binary indicator. Instead, we define the effective zone fire access as:

$$\text{zone_fire_access}[i] = 0.5 + 0.5 * \text{fire_unit_access}[i]$$

This formulation assigns a baseline access level of 0.5 to every zone, reflecting the passive fire suppression capacity of park rangers and existing infrastructure even in the absence of a dedicated fire unit. The remaining 0.5 is contributed by the fire unit access variable. This design choice ensures that no zone is treated as entirely without response capacity, which would be unrealistic given the park's permanent ranger presence, while still meaningfully rewarding the placement of dedicated fire units.

As in the poaching stage, overall fire prevention in zone i requires both detection and access to occur simultaneously. The product of $\text{zone_fire_detection}[i]$ and $\text{zone_fire_access}[i]$ is again linearized using McCormick envelope constraints:

$$\begin{aligned} \text{fire_prevent_zone}[i] &\leq \text{zone_fire_detection}[i]; \text{fire_prevent_zone}[i] \leq \text{zone_fire_access}[i]; \\ \text{fire_prevent_zone}[i] &\geq \text{zone_fire_detection}[i] + \text{zone_fire_access}[i] - 1 \end{aligned}$$

Similar to the poaching stage, a fire can only be put out if it is discovered and a response unit can get to it. In order to ensure MIP-compatibility and accurately depict the joint dependence of prevention on both surveillance and intervention capacity, the McCormick linearization maintains this logical interaction within a fully linear model.

After optimising this model in CPLEX we got the following placement:

```
fire_unit = [0 0 0 1 0 0 0 0 0 0 0 0 0 0];
fire_drones = [0 0 0 12 12 0 12 0 0 0 0 0 12 0 0];
```

3. Protection over time and staffing need

With 150 deployed drones, 60 patrols and 1 fire unit deployed according to the model results Etosha National Park will have a weighted value of protection level equal to 9752 for fire events and 231849.794 for poaching. Overall, the unweighted protection level with this equipment will be 0.7. If the officials would like to increase the protection level, they may increase the patrol staff and number of detecting drones and calculate new optimal positioning using our model.

4. Sensitivity and scenario analysis

We perform a sensitivity analysis by methodically altering important model parameters and analyzing their impact on the deployment strategy and objective values in order to evaluate the robustness of the optimized allocation.

When compared to similar reductions in drone resources, the effectiveness of prevention is disproportionately affected when the budget for ground patrols is cut. This is because patrols are the sole driver of the Access variable, meaning that detection and response capacity are not substitutable: a zone that is well-monitored by drones but poorly staffed on the ground will consistently underperform, as interception requires physical presence regardless of surveillance quality.

Drone budget reductions affect zones unevenly. Road-inaccessible zones, where ground patrols contribute nothing to coverage, depend entirely on aerial surveillance and are therefore the most vulnerable component of the network under any resource constraint. These zones experience a sharper degradation in prevention than road-accessible zones, where patrols partially compensate for reduced drone presence.

The two-stage structure of the model introduces an important compounding effect: any reduction in poaching-stage resources also weakens wildfire detection, since pre-existing patrols and drones feed directly into fire coverage as fixed parameters in the second stage. This interdependence means that resource decisions made in the first stage carry consequences that extend beyond poaching prevention alone.

Lastly, a certain amount of natural resilience is offered by the spatial spillover mechanism. The contributions from nearby zones partially offset moderate patrol number reductions, resulting in a smoother decline in prevention than would be expected from a zone-by-zone analysis. This buffer is

constrained, though, and no matter how centrally located resources are allocated, peripheral zones with few neighbors continue to be structurally exposed.

5. Adaptation to other protected areas

We also checked how well our model can be applied to Yellowstone national park, USA and Komodo national park, Indonesia. The parks were chosen to be on different continents, as we assume that geographical position influences our model the most, as well as the prevalence of certain types of species.

6.1. Poaching protection

Yellowstone national park

The main problem related to poaching in Yellowstone national park is wolf poaching and grizzly bears are also important to protect from such acts, as they are endangered.

So, in adaptation of the poaching model the park will place wolves and grizzly bears instead of rhino level. The habitats of these species will also define division of the reserve into zones – the zones in Etosha model depended mostly on rhino population levels across the park. Hence, in the new circumstances we also will base zone division on animal prevalence in different regions of the park.

The areas of the regions, fencing parameters will be identified using the map from Yellowstone park website:

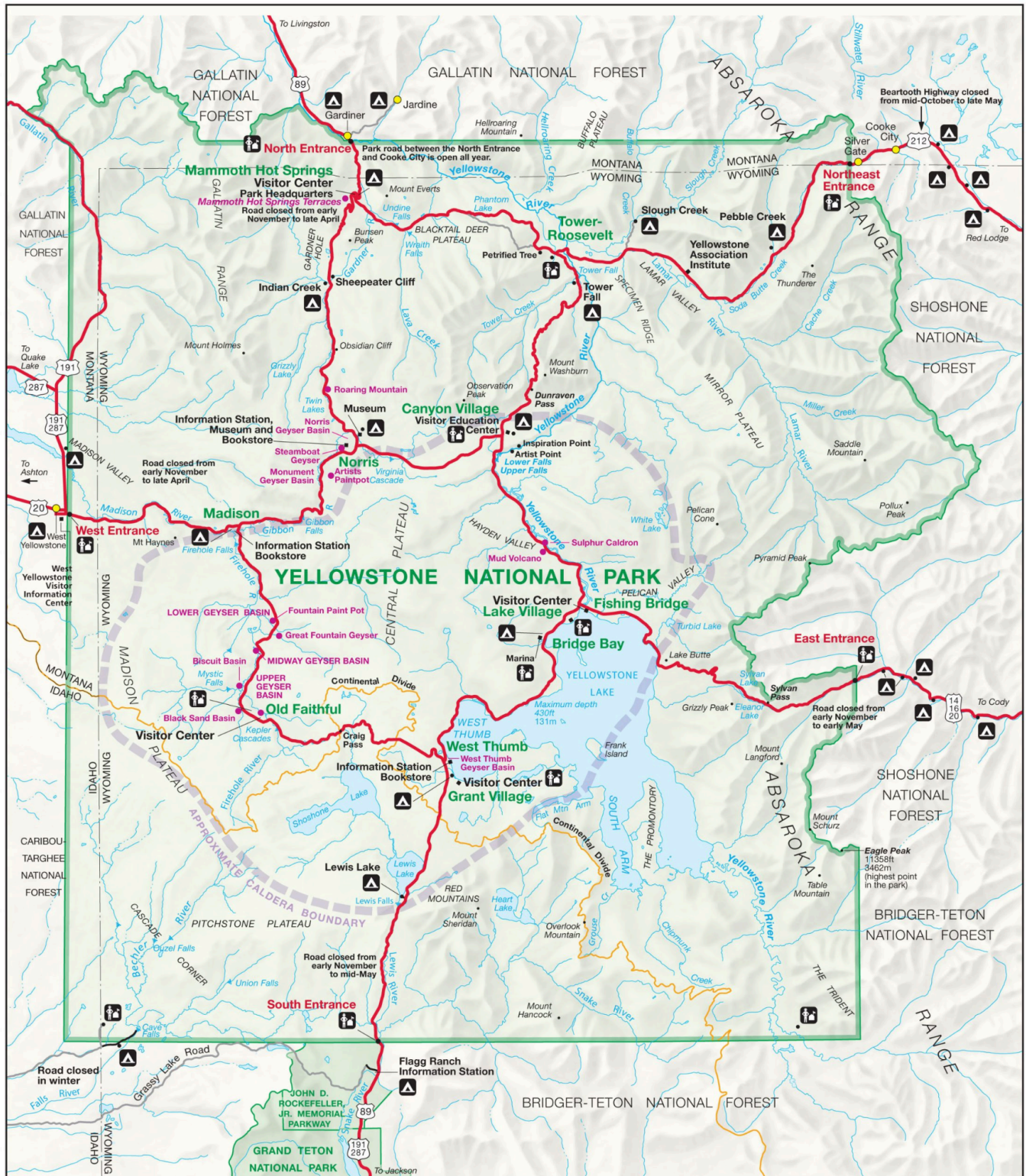


Figure 3: Yellowstone national park map

Instead of waterholes it will be important to consider lakes and rivers instead of waterholes, as they are distinguishing features in this park. (Introduce variables lake_area, river_length in the zone)

Komodo national park

In Komodo national park Timor deer and Komodo dragons are the most vulnerable species in terms of poaching.

So, in adaptation of the poaching model the park will place Timodo deers and Komodo dragons instead of rhino level. The habitats of these species will also define division of the reserve into zones – the zones in Etosha model depended mostly on rhino population levels across the park. Hence, in the new circumstances we also will base zone division on animal prevalence in different regions of the park.

The areas of the regions, fencing parameters will be identified using the map from Komodo park website:



Figure 4: Komodo national park tourism map

It will also require further research into water objects located in the park.

6.2. Wildfire protection

For both regions we should calculate rain level and number of wildfires throughout the last years, using the information available in articles and research papers.

6.3. Modification of parametres

Apart from finding the new values for all the variables used: area, road network, presence of the main camp, species priorities and their habitats, we should add some other threat types relevant for this particular park.

For instance, trash issues on the main roads of Yellowstone and plastic pollution to the coral reefs in Komodo. This unique problem would require the creation of the same model using the same principle, as we used for poach and fire modelling.

6. Conclusion and discussion

7.1. Model evaluation

Strengths

- Our model suggests a quantitative measure of protection level
- Our model account for both natural environment and man-made infrastructure
- We allocate various types of resources: patrol units, drones, fire stations
- Our model is flexible in terms of resources limitation; the number of units available can be easily plugged in the formula
- Our model can be adjusted to other contexts, as was shown in the previous section

Limitations

- To apply the model to the unknown context it is necessary to analyze the basic problems of the park and sort them by importance, assess the importance of preserving each of the preexisting animal species, to strengthen the protection of rare ones. Our model would benefit from a standardized structure for such an application.
- Although we our model consider multiple factors there is still a room for improvement: it could also consider season, as it affects wildfire probability (it is higher in dry season – June-December and lower in rain season – October-May)

7.2. Conclusion

The objective of this paper is to plan the protection of the natural park, by resource allocation using a mathematical model. We achieved this goal, as our model allowed us to allocate protection resources in an effective way. We identified where patrols, drones and fire units should be placed on the map below:

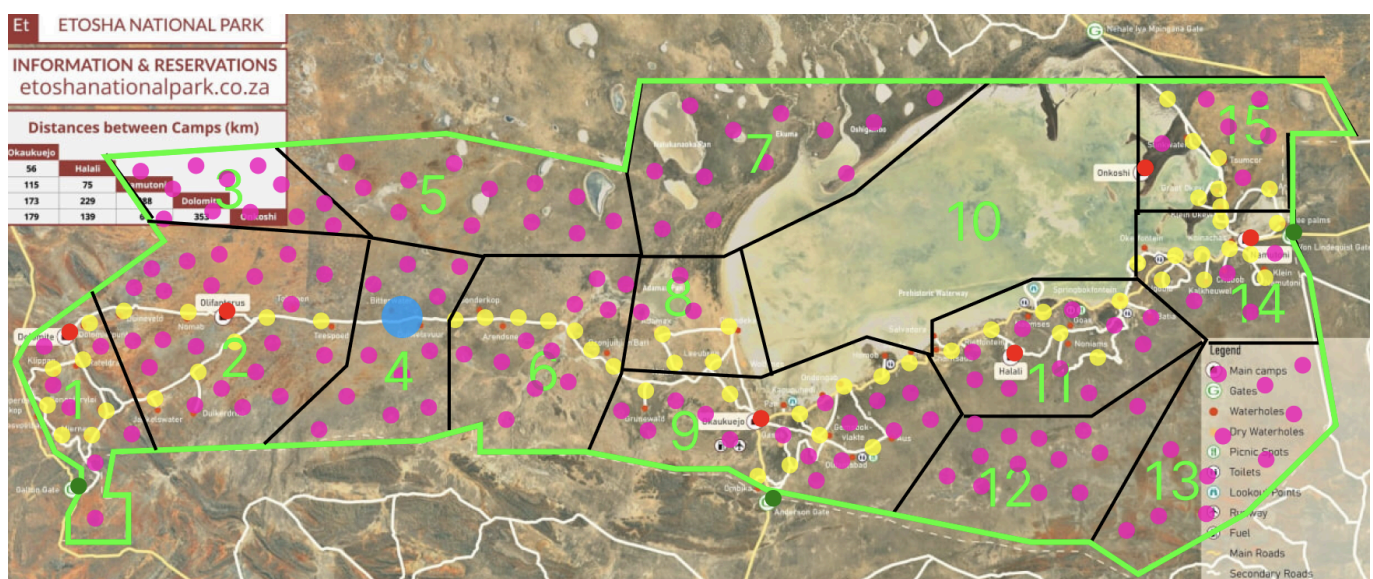


Figure 5: Map with suggested protection resources allocation. Drones are represented with pink, patrols are yellow, fire unit is blue.

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Appendices

Appendix A: Poaching protection model OPL code

```
int N = 15;
range Z = 1..N;

float area[Z] = [
1408,3219,1207,1811,2213,
2011,2625,905,2715,4325,
1609,1710,2011,1257,1106
];
float road[Z] = [
0.9,0.7,0,0.3,0,
0.6,0,0.5,0.8,0,
0.75,0,0,0.8,0.6
```

```

];
float water[Z] = [
0.7,0.5,0,0.2,0,
0.3,0,0.1,0.9,0,
0.8,0,0,0.7,0.5
];
float rhino[Z] = [
2,2,1,0,0,
2,0,0,2,0,
2,2,0,2,2
];
int animal[Z] = [
2,2,1,1,0,
2,0,2,2,0,
2,2,0,2,2
];
int maincamp[Z] = [
1,1,0,0,0,
0,0,0,1,0,
1,0,0,1,1
];

int neigh[Z][Z] = [
[0,1,1,1,0,0,0,0,0,0,0,0,0,0],
[1,0,0,0,1,0,0,0,0,0,0,0,0,0],
[1,0,0,1,0,1,0,0,0,0,0,0,0,0],
[1,0,1,0,1,1,1,0,0,0,0,0,0,0],
[0,1,0,1,0,0,0,1,1,0,0,0,0,0],
[0,0,1,1,0,0,1,0,1,1,0,0,0,0],
[0,0,0,1,0,1,0,1,0,0,1,1,0,0],
[0,0,0,1,0,1,0,1,0,0,1,1,0,0],
[0,0,0,0,1,0,1,0,0,1,0,0,0,0],
[0,0,0,0,1,1,0,0,0,1,1,0,0,1],
[0,0,0,0,0,1,0,1,1,0,0,1,0,1],
[0,0,0,0,0,0,1,0,1,0,0,0,1,0],
[0,0,0,0,0,0,1,0,1,0,0,0,1,0],
[0,0,0,0,0,0,1,0,0,1,0,0,0,1],
[0,0,0,0,0,0,0,0,0,0,1,0,0,0],
[0,0,0,0,0,0,0,0,1,1,0,1,0,0],
[0,0,0,0,0,0,0,0,1,0,0,0,1,1],
[0,0,0,0,0,0,0,0,1,0,0,0,1,0]
];

float patrolRange = 80;
float droneRange = 128;

float beta1 = 0.6;
float beta2 = 0.3;
float cp = 0.5;

float W1 = 1;
float W2 = 3;
float W3 = 0.9;

int maxPatrolPerZone = 11;

int nbNeigh[i in Z] = sum(j in Z) neigh[i][j];

float alpha[i in Z] =
(nbNeigh[i] > 0) ? 1.0 / nbNeigh[i] : 0;

float poachRisk[i in Z] =
(W2 * rhino[i] + W1 * animal[i])
* (road[i] + 1)
* (water[i] + 1)
* (2 - W3 * maincamp[i]);

```



```

dvar int+ P[Z];
dvar int+ D[Z];
dvar float Coverage[Z] in 0..1;
dvar float Det[Z] in 0..1;
dvar float Access[Z] in 0..1;
dvar float Prevent[Z] in 0..1;

minimize
  sum(i in Z)
    poachRisk[i] * (1 - Prevent[i]) * area[i];

subject to {

  sum(i in Z) P[i] == 60;
  sum(i in Z) D[i] == 100;

  forall(i in Z)
    P[i] <= maxPatrolPerZone;

  forall(i in Z) {
    Coverage[i] == (patrolRange * road[i] * P[i]
      + droneRange * D[i]) / area[i];
    Coverage[i] <= 1;
    Coverage[i] >= 0;
  }

  forall(i in Z) {
    Det[i] <= beta1 * Coverage[i];
    Det[i] <= beta1 * cp + beta2 * (Coverage[i] - cp);
    Det[i] <= 1;
  }

  forall(i in Z){
    Access[i] <= P[i] / maxPatrolPerZone
      + alpha[i] *
        sum(j in Z : neigh[i][j] == 1)
          P[j] / maxPatrolPerZone;
    Access[i] <= 1;
    Access[i] >= 0;
  }

  forall(i in Z){
    Prevent[i] <= Det[i];
    Prevent[i] <= Access[i];
    Prevent[i] >= Det[i] + Access[i] - 1;
  }
}

```

Appendix B: Optimization results

```

// solution (integer optimal, tolerance) with objective 231849.794
// Quality Incumbent solution:
// MILP objective                2.3184979400e+05
// MILP solution norm |x| (Total, Max)    1.81854e+02  2.20000e+01
// MILP solution error (Ax=b) (Total, Max)  1.09635e-14  6.21725e-15
// MILP x bound error (Total, Max)        0.00000e+00  0.00000e+00
// MILP x integrality error (Total, Max)   1.91624e-13  6.39488e-14
// MILP slack bound error (Total, Max)     4.16334e-16  1.11022e-16
//
Prevent = [0.4483
  0.44897 0.34089 0 0 0.44836 0 0.33033 0.44702 0 0.44459 0.44193 0

```

```

    0.44021 0.44946];
P = [7 7 0 0 0 7 1 4 10 0 5 1 0 11 7];
D = [7 22 6 0 0 13 0 3 16 0 10 13 0 4 6];
Coverage = [0.99432 0.99658 0.63629 0 0 0.99453 0 0.6011 0.99006 0 0.98198 0.9731
    0 0.96738 0.99819];
Det = [0.4483 0.44897 0.34089 0 0 0.44836 0 0.33033 0.44702 0 0.44459 0.44193
    0 0.44021 0.44946];
Access = [0.4483 0.44897 0.34089 0 0 0.44836 0 0.33033 0.44702 0 0.44459 0.44193
    0 0.44021 0.44946];

```

Appendix C: Wildfire prevention model

```

int N = 15;
range Z = 1..N;
float area[Z] = [
1408,3219,1207,1811,2213,
2011,2625,905,2715,4325,
1609,1710,2011,1257,1106
];
float road[Z] = [
0.9,0.7,0,0.3,0,
0.6,0,0.5,0.8,0,
0.75,0,0,0.8,0.6
];
float rain[Z] = [
0.2,0.1,0.3,0.1,0.2,
0.25,0.15,0.05,0.2,0.1,
0.15,0.2,0.1,0.1,0.05
];
float fire_accidents[Z] = [
2,1,0,3,1,
2,1,0,2,0,
1,2,0,1,0
];
int const_patrols[Z] = [7, 7, 0, 0, 0, 7, 1, 4, 10, 0, 5, 1, 0, 11, 7];
int const_drones[Z] = [7, 22, 6, 0, 0, 13, 0, 3, 16, 0, 10, 13, 0, 4, 6];
int neigh[Z][Z] = [
[0,1,1,1,0,0,0,0,0,0,0,0,0,0,0],
[1,0,0,0,1,0,0,0,0,0,0,0,0,0,0],
[1,0,0,1,0,1,0,0,0,0,0,0,0,0,0],
[1,0,1,0,1,1,1,0,0,0,0,0,0,0,0],
[0,1,0,1,0,0,0,1,1,0,0,0,0,0,0],
[0,0,1,1,0,0,1,0,1,1,0,0,0,0,0],
[0,0,0,1,0,1,0,1,0,0,1,1,0,0,0],
[0,0,0,0,1,0,1,0,0,1,0,0,0,0,0],
[0,0,0,0,1,1,0,0,0,1,1,0,0,1,1],
[0,0,0,0,0,1,0,1,1,0,0,1,0,1,0],
[0,0,0,0,0,0,1,0,1,0,0,0,1,0,0],
[0,0,0,0,0,0,1,0,0,1,0,0,0,1,0],
[0,0,0,0,0,0,0,0,0,0,1,0,0,0,1],
[0,0,0,0,0,0,0,0,1,1,0,1,0,0,1],
[0,0,0,0,0,0,0,0,1,0,0,0,1,1,0]
];
float patrol_range = 80;
float drone_range = 128;
int max_fire_unit = 1;
int max_drones = 60;
int maxDronesPerZone = 12;
int nbNeigh[i in Z] = sum(j in Z) neigh[i][j];
float alpha[i in Z] = (nbNeigh[i] > 0) ? 1.0 / nbNeigh[i] : 0;

```

```

dvar int+ fire_unit[Z] in 0..1;
dvar int+ fire_drones[Z] in 0..50;
dvar float fire_unit_access[Z] in 0..1;
dvar float zone_fire_access[Z] in 0..1;
dvar float const_drone_coverage[Z] in 0..1e6;
dvar float const_patrol_coverage[Z] in 0..1e6;
dvar float adj_drone_coverage[Z] in 0..1e6;
dvar float zone_fire_detection[Z] in 0..1;
dvar float fire_prevent_zone[Z] in 0..1;

minimize
sum(i in Z)
  (1 - rain[i]) * fire_accidents[i] * (1 - fire_prevent_zone[i]) * area[i];

subject to {

sum(i in Z) fire_unit[i] <= max_fire_unit;
sum(i in Z) fire_drones[i] <= max_drones;

forall(i in Z)
  fire_drones[i] <= maxDronesPerZone;

forall(i in Z){
  fire_unit_access[i] <= fire_unit[i] + sum(j in Z:neigh[i][j]==1) alpha[i]*fire_unit[j];
  fire_unit_access[i] >= fire_unit[i];
}

forall(i in Z){
  zone_fire_access[i] == 0.5 + 0.5 * fire_unit_access[i];
}

forall(i in Z){
  const_drone_coverage[i] == drone_range * const_drones[i];
  const_patrol_coverage[i] == road[i] * patrol_range * const_patrols[i];
}

forall(i in Z){
  adj_drone_coverage[i] == drone_range * fire_drones[i];
}

forall(i in Z){
  zone_fire_detection[i] <= adj_drone_coverage[i] + const_drone_coverage[i] + const_patrol_coverage[i];
  zone_fire_detection[i] <= 1;
  zone_fire_detection[i] >= 0;
}

forall(i in Z){
  fire_prevent_zone[i] <= zone_fire_detection[i];
  fire_prevent_zone[i] <= zone_fire_access[i];
  fire_prevent_zone[i] >= zone_fire_detection[i] + zone_fire_access[i] - 1;
}
}

```

Appendix D: Optimisation results

```

// solution (optimal) with objective 9751.95833333333
// Quality Incumbent solution:
// MILP objective          9.7519583333e+03
// MILP solution norm |x| (Total, Max)    2.39355e+04 2.81600e+03
// MILP solution error (Ax=b) (Total, Max) 2.57683e-13 9.09273e-14
// MILP x bound error (Total, Max)      0.00000e+00 0.00000e+00

```

```
// MILP x integrality error (Total, Max)    0.00000e+00 0.00000e+00
// MILP slack bound error (Total, Max)    0.00000e+00 0.00000e+00
//
fire_prevent_zone = [0.66667
    0.5 0 1 0.625 0.6 0.6 0 0.5 0 0.5 0.5 0 0.5 0];
fire_unit = [0 0 0 1 0 0 0 0 0 0 0 0 0 0];
fire_drones = [0 0 12 12 12 0 12 0 0 0 0 0 12 0 0];
fire_unit_access = [0.33333 0 0 1 0.25 0.2 0.2 0 0 0 0 0 0 0];
zone_fire_access = [0.66667 0.5 0.5 1 0.625 0.6 0.6 0.5 0.5 0.5 0.5 0.5 0.5
    0.5 0.5];
const_drone_coverage = [896 2816 768 0 0 1664 0 384 2048 0 1280 1664 0 512 768];
const_patrol_coverage = [504 392 0 0 0 336 0 160 640 0 300 0 0 704 336];
adj_drone_coverage = [0 0 0 1536 1536 0 1536 0 0 1536 0 0 1536 0 0];
zone_fire_detection = [0.66667 0.5 0 1 0.625 0.6 0.6 0 0.5 0 0.5 0.5 0 0.5 0];
```